We now have *separate* equations for **E** and **B**, but they are of *second* order; that's the price you pay for decoupling them.

In vacuum, then, each Cartesian component of **E** and **B** satisfies the **three-dimensional wave equation**,

$$\nabla^2 f = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}.$$

(This is the same as Eq. 9.2, except that $\partial^2 f/\partial z^2$ is replaced by its natural generalization, $\nabla^2 f$.) So Maxwell's equations imply that empty space supports the propagation of electromagnetic waves, traveling at velocity

$$v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3.00 \times 10^8 \text{ m/s}, \tag{9.42}$$

which happens to be precisely the speed of light, c. Evidently,

$$\boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}.} \tag{9.43}$$

The implication is astounding: perhaps light *is* an electromagnetic wave.[6] Of course, this conclusion does not surprise anyone today, but imagine what a revelation it must have been in Maxwell's time! Remember how ϵ_0 and μ_0 came into the theory in the first place: they were constants in Coulomb's law and the Biot–Savart law, respectively. You measure them in experiments involving charged pith balls, batteries, and wires – experiments having nothing whatever to do with light. And yet, according to Maxwell's theory, you can calculate c from these two numbers. Notice the crucial role played by Maxwell's contribution to Ampère's law ($\mu_0\epsilon_0\partial\mathbf{E}/\partial t$): without it, the wave equation would not emerge, and there would be no electromagnetic theory of light.

9.2.2 Monochromatic Plane Waves

For reasons discussed in Section 9.1.2, we may confine our attention to sinusoidal waves of frequency ω. Since different frequencies in the visible range correspond to different *colors*, such waves are called **monochromatic** (Table 9.1). Suppose, moreover, that the waves are traveling in the z direction and have no x or y dependence; these are called **plane waves**,[7] because the fields are uniform over every plane perpendicular to the direction of propagation (Fig. 9.9). We are interested, then, in fields of the form

[6] As Maxwell himself put it, "We can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena." See I. Tolstoy, *James Clerk Maxwell, A Biography*, University of Chicago Press, Chicago (1983).

[7] For a discussion of *spherical* waves, at this level, see J. R. Reitz, F. J. Milford, and R. W. Christy, *Foundations of Electromagnetic Theory*, 3rd ed., Addison-Wesley, Reading, MA (1979), Sect. 17-5. Or work Prob. 9.37. Of course, over small enough regions *any* wave is essentially plane, as long as the wavelength is much less than the radius of the curvature of the wave front.

Table 9.1.

The Electromagnetic Spectrum

Frequency (Hz)	Type	Wavelength (m)
10^{22}		10^{-13}
10^{21}	Gamma rays	10^{-12}
10^{20}		10^{-11}
10^{19}		10^{-10}
10^{18}	X rays	10^{-9}
10^{17}		10^{-8}
10^{16}	Ultraviolet	10^{-7}
10^{15}	Visible	10^{-6}
10^{14}	Infrared	10^{-5}
10^{13}		10^{-4}
10^{12}		10^{-3}
10^{11}		10^{-2}
10^{10}	Microwave	10^{-1}
10^{9}		1
10^{8}	TV, FM	10
10^{7}		10^{2}
10^{6}	AM	10^{3}
10^{5}		10^{4}
10^{4}	RF	10^{5}
10^{3}		10^{6}

The Visible Range

Frequency (Hz)	Color	Wavelength (m)
1.0×10^{15}	Near ultraviolet	3.0×10^{-7}
7.5×10^{14}	Shortest visible blue	4.0×10^{-7}
6.5×10^{14}	Blue	4.6×10^{-7}
5.6×10^{14}	Green	5.4×10^{-7}
5.1×10^{14}	Yellow	5.9×10^{-7}
4.9×10^{14}	Orange	6.1×10^{-7}
3.9×10^{14}	Longest visible red	7.6×10^{-7}
3.0×10^{14}	Near infrared	1.0×10^{-6}

$$\tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{i(kz-\omega t)}, \quad \tilde{\mathbf{B}}(z,t) = \tilde{\mathbf{B}}_0 e^{i(kz-\omega t)}, \tag{9.44}$$

where $\tilde{\mathbf{E}}_0$ and $\tilde{\mathbf{B}}_0$ are the (complex) amplitudes (the *physical* fields, of course, are the real parts of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{B}}$), and $\omega = ck$.

Now, the wave equations for $\mathbf{E}$ and $\mathbf{B}$ (Eq. 9.41) were derived from Maxwell's equations. However, whereas every solution to Maxwell's equations (in empty space) must obey the wave equation, the converse is *not* true; Maxwell's equations impose extra constraints on $\tilde{\mathbf{E}}_0$ and $\tilde{\mathbf{B}}_0$. In particular, since $\boldsymbol{\nabla} \cdot \mathbf{E} = 0$ and $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$, it follows[8] that

[8] Because the real part of $\tilde{\mathbf{E}}$ differs from the imaginary part only in the replacement of sine by cosine, if the former obeys Maxwell's equations, so does the latter, and hence $\tilde{\mathbf{E}}$ as well.

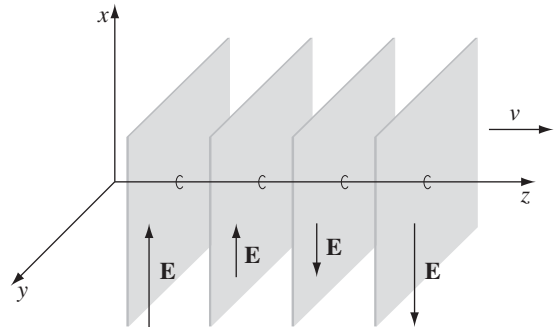


Fig. 9.9

$$(\tilde{E}_0)_z = (\tilde{B}_0)_z = 0. \tag{9.45}$$

That is, *electromagnetic waves are transverse*: the electric and magnetic fields are perpendicular to the direction of propagation. Moreover, Faraday's law, $\boldsymbol{\nabla} \times \mathbf{E} = -\partial \mathbf{B}/\partial t$, implies a relation between the electric and magnetic amplitudes, to wit,

$$-k(\tilde{E}_0)_y = \omega(\tilde{B}_0)_x, \quad k(\tilde{E}_0)_x = \omega(\tilde{B}_0)_y, \tag{9.46}$$

or, more compactly,

$$\tilde{\mathbf{B}}_0 = \frac{k}{\omega}(\hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0). \tag{9.47}$$

Evidently, **E** and **B** are *mutually perpendicular*, and perpendicular to the direction of propagation (here, z), so they really have only one (nonzero) component each; they are *in phase*, and their (real) amplitudes are related by

$$B_0 = \frac{k}{\omega}E_0 = \frac{1}{c}E_0. \tag{9.48}$$

The fourth of Maxwell's equations, $\boldsymbol{\nabla} \times \mathbf{B} = \mu_0\epsilon_0(\partial \mathbf{E}/\partial t)$, does not yield an independent condition; it simply reproduces Eq. 9.46.

Example 9.2. If **E** points in the x direction, then **B** points in the y direction (Eq. 9.47):

$$\tilde{\mathbf{E}}(z,t) = \tilde{E}_0 e^{i(kz-\omega t)}\hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}(z,t) = \frac{1}{c}\tilde{E}_0 e^{i(kz-\omega t)}\hat{\mathbf{y}},$$

or (taking the real part)

$$\boxed{\mathbf{E}(z,t) = E_0\cos(kz-\omega t+\delta)\hat{\mathbf{x}}, \quad \mathbf{B}(z,t) = \frac{1}{c}E_0\cos(kz-\omega t+\delta)\hat{\mathbf{y}}.} \tag{9.49}$$

This is the paradigm for a monochromatic plane wave (see Fig. 9.10). The wave as a whole is said to be polarized in the x direction (by convention, we use the direction of **E** to specify the polarization of an electromagnetic wave).

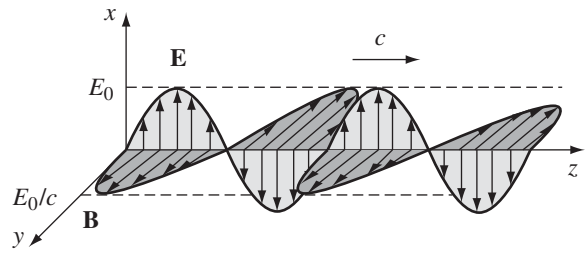


Fig. 9.10

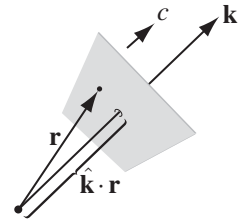


Fig. 9.11

There is nothing special about the z direction, of course – we can easily generalize to monochromatic plane waves traveling in an arbitrary direction. The notation is facilitated by the introduction of the **propagation** (or **wave**) **vector**, $\mathbf{k}$, pointing in the direction of propagation, whose magnitude is the wave number k. The scalar product $\mathbf{k} \cdot \mathbf{r}$ is the appropriate generalization of kz (Fig. 9.11), so

$$\boxed{\begin{aligned}\tilde{\mathbf{E}}(\mathbf{r}, t) &= \tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\hat{\mathbf{n}},\\ \tilde{\mathbf{B}}(\mathbf{r}, t) &= \frac{1}{c}\tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}(\hat{\mathbf{k}} \times \hat{\mathbf{n}}) = \frac{1}{c}\hat{\mathbf{k}} \times \tilde{\mathbf{E}},\end{aligned}} \tag{9.50}$$

where $\hat{\mathbf{n}}$ is the polarization vector. Because $\mathbf{E}$ is transverse,

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = 0. \tag{9.51}$$

(The transversality of $\mathbf{B}$ follows automatically from Eq. 9.50.) The actual (real) electric and magnetic fields in a monochromatic plane wave with propagation vector $\mathbf{k}$ and polarization $\hat{\mathbf{n}}$ are

$$\mathbf{E}(\mathbf{r}, t) = E_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta)\hat{\mathbf{n}}, \tag{9.52}$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c}E_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta)(\hat{\mathbf{k}} \times \hat{\mathbf{n}}). \tag{9.53}$$

Problem 9.9. Write down the (real) electric and magnetic fields for a monochromatic plane wave of amplitude E_0, frequency ω, and phase angle zero that is (a) traveling in the negative x direction and polarized in the z direction, and (b) traveling in the direction from the origin to the point $(1, 1, 1)$, with polarization parallel to the xz plane. In each case, sketch the wave, and give the explicit Cartesian components of $\mathbf{k}$ and $\hat{\mathbf{n}}$.